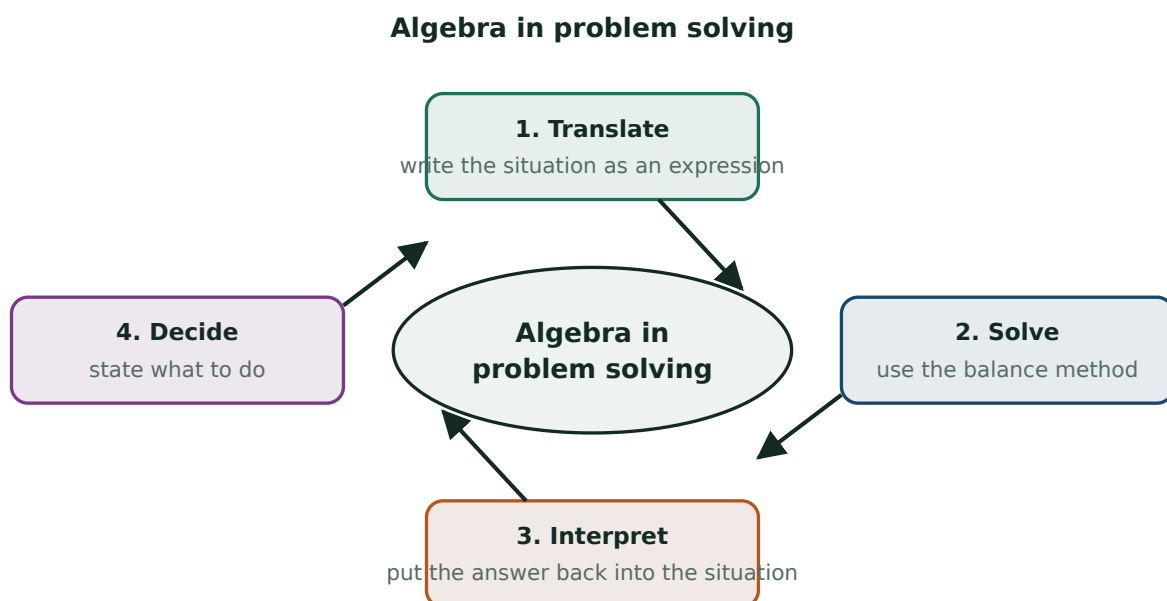


Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Topic 3.3 - Using algebra in problem solving

A real classroom situation



Unit 1's four-step strategy, with algebra as the plan step's sharpest tool.

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Two tailors in the same bazaar post different rate cards. Tailor A: "500 rupees per suit, plus 200 for buttons, however many suits." Tailor B: "600 rupees per suit, buttons free." A trainee, Amjad, needs one suit with buttons and cannot decide. His friend laughs: "Just try both - A is 700, B is 600, take B." But Amjad's teacher asks a better question: "What if the school orders suits for the whole cricket team?" With one suit B wins, but the two situations follow two rules: A

costs $500n + 200$ and B costs $600n$, where n is the number of suits. Setting the rules side by side - which is all algebra is doing here - shows that at $n = 2$ they tie at 1,200 rupees, and from $n = 3$ onward A is cheaper, because A grows by 500 per suit while B grows by 600. One comparison of two expressions settled a whole family of questions at once.

Explanation

Use of algebra in problem solving

The everyday phrase	The algebra it becomes
a fixed amount plus a rate per item: "60 rupees to start, then 15 per kilometre"	$60 + 15k$
the total splits into equal groups: "240 pupils placed in groups of 15"	$240 / 15 = g$
one quantity is so much more than another: "the van carries 12 more than the bus"	$v = b + 12$
the total is a rate times a quantity: "notebooks at 80 rupees each"	$\text{cost} = 80n$
from what amount onwards: "when does Printer B stop being cheaper?"	$2,000 + 15n = 1,200 + 25n$

Read the situation, pick the shape, write the rule.

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Algebra earns its place in problem solving by **compressing a whole family of situations into one rule**. The method is a loop, shown in the concept map above: translate the situation into an expression or equation, solve with the balance method from Topic 3.2, interpret the result back into the situation, and decide. It is the four-step strategy of Unit 1 with algebra as the plan step's sharpest tool.

The table above collects the five translation patterns that cover most daily problems. Study the right-hand column: each everyday phrase has exactly one algebraic shape.

Worked example. A tutor charges 400 rupees per session plus 250 for materials. A family paid 2,650 rupees in total. How many sessions did they take?

1. **Translate:** $\text{cost} = 250 + 400s$, and the cost was 2,650, so $250 + 400s = 2,650$.
2. **Solve:** subtract 250 from both sides: $400s = 2,400$; divide by 400: $s = 6$.
3. **Interpret:** s counts sessions, so the family took 6 sessions.
4. **Decide and check:** $250 + 400 \times 6 = 250 + 2,400 = 2,650$. True.

A common misconception to correct. Trainees who fear letters treat algebra as a subject separate from "real maths". The opposite is true: algebra is what makes daily arithmetic reusable. The tailor's rate card, the van fare, the tutor's bill and the stationery budget are all the same equation wearing different clothes. Once you see the shape, every new instance is already solved. Marecek, Anthony-Smith and Mathis (2020) build their algebra chapters on exactly this translation skill, from "the language of algebra" to solving applied problems.

Activity: Rate card clinic

Work in pairs. Each pair writes two rate cards from real life (van fares, photocopy charges, canteen prices with a fixed part, mobile packages with a monthly fee plus per-minute rate). Swap cards with another pair. For each card: write the cost expression, compute the cost for 3, 10 and 25 units, and state the smallest number of units for which the card beats a flat rate of 500 rupees. Defend one answer to the class. About 25 minutes.

Check your understanding

1. Write the cost expression for a photocopy shop that charges 30 rupees to open the machine plus 8 rupees per copy, and compute the cost of 25 copies. (*Apply*)
2. Translate into an equation and solve: "A family paid 2,650 rupees to a tutor who charges 250 rupees for materials plus 400 per session. How many sessions?" (*Apply*)

3. A van costs 500 rupees plus 30 per kilometre; a rival costs 300 rupees plus 40 per kilometre. Write both cost expressions, and find the distance at which they cost the same. (*Apply*)
4. In your own words, explain why comparing two expressions answers a whole family of questions at once, using the tailor example. (*Understand*)
5. A trainee solves $250 + 400s = 2,650$ as $s = 2,650 - 250 - 400 = 2,000$. Identify the error and state the correct first move. (*Analyze*)

Summary

Algebra compresses a family of situations into one rule: translate the situation into an expression or equation, solve with the balance method, interpret the result back into the situation, and decide. Five translation patterns cover most daily problems, and comparing two expressions settles questions the arithmetic of a single case cannot even ask.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can translate a fare, budget or rate situation into an expression or equation.
- ☐ I can solve the equation and interpret the answer back into the situation, with a check.
- ☐ I can compare two pricing rules by comparing their expressions.

Try this at your practicum school

Collect one real rate card or price rule from the school or its bazaar (photocopy shop, van, canteen). Write its cost expression, compute the cost for three realistic quantities, and ask the shopkeeper whether your numbers match their bill. Note any difference and its cause.

Summative task

A school event needs printed programmes. Printer A charges 2,000 rupees setup plus 15 rupees per programme; Printer B charges 1,200 rupees setup plus 25 rupees per programme. The school needs 150 programmes. In a short written answer: (a) write both cost expressions, (b) compute both totals for 150 programmes and state which is cheaper, and (c) find the number of programmes at which both printers cost the same, and recommend which printer the school should choose if the print run might grow or shrink, with a reason.

Mini-rubric (each criterion 0-2 marks):

- *Expressions:* $A = 2,000 + 15n$ and $B = 1,200 + 25n$ (variables named).
- *Totals and comparison:* A costs 4,250 and B costs 4,950 for 150, so A is cheaper by 700 rupees.
- *Break-even and recommendation:* $2,000 + 15n = 1,200 + 25n$ gives $800 = 10n$, so $n = 80$; the recommendation weighs the break-even against the chance of a smaller run (below 80, B is cheaper) with a reason. **(Analyze)**

Further reading

- Marecek, L., Anthony-Smith, M., & Mathis, M. H. (2020). *Prealgebra 2e* (Ch. 2 The Language of Algebra; Ch. 8 Solving Linear Equations). OpenStax.
<https://openstax.org/details/books/prealgebra-2e>
- National Curriculum Council, Ministry of Federal Education and Professional Training, Government of Pakistan. *Mathematics: Suggested Guidelines (Grades 1-8)*.
<https://ncc.gov.pk/Detail/ZjgzYzg2MmMtZDc0Zi00NjEzLTk5ZmYtZGJiNjc5ODIjOGUx>

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